When Trusted Constraints Help and Hurt in Offline-to-Online Reinforcement Learning

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Living draft, 2026-05-13

Abstract

Offline-to-online reinforcement learning can reuse logged experience before interacting with the environment, but its benefit depends on the quality of the offline replay data and on how strongly the algorithm constrains its policy. We study a focused D4RL MuJoCo replay setting through three design families: value conservatism, normal or weakly regularized online learning, and policy/behavior regularization. The current draft is anchored by completed C-line experiments. Plain TD3+BC is weak on `hopper-medium-replay-v2`: it reaches 22.43 normalized score after 50k offline steps, while ReBRAC-lite reaches 34.48 and SSAR reaches 38.56 final / 43.97 best under the same 50k screen. Mechanism experiments suggest that SSAR's IQL-qv trusted-action selection is a key driver: removing trusted selection collapses performance, while an aligned teacher-label selector, ATLAS, reaches 69.97 and 68.11 final score on hopper seed0/seed1 at 100k steps. Shuffling labels collapses ATLAS to 18.78, indicating that label alignment matters rather than arbitrary weighting. However, a minimal offline-to-online slice reveals the central limitation: TD3+BC improves from 22.43 to 39.64 after 10k online steps, whereas ATLAS loses its offline advantage under the current online regularization. These results support a constraint-transfer view: trusted-action regularization can improve offline learning under low-quality data, but the same constraint must be released or reweighted during online adaptation.

1 Introduction

Offline-to-online reinforcement learning (RL) promises to reduce online sample cost by first learning from a fixed replay dataset and then fine-tuning through environment interaction. This setup is attractive for course-project-scale empirical work because it exposes a concrete tradeoff: the offline dataset can provide a useful initialization, but low-quality replay data can also bias or constrain the learned policy.

The central question in this project is:

> Under low-quality replay data, how do different forms of conservatism and regularization affect offline-to-online fine-tuning?

We organize the project into three tracks. A-line studies value conservatism, such as CQL-family methods. B-line provides a normal or non-conservative contrast, such as PPO, SAC, or vanilla TD3-style online fine-tuning. C-line studies policy and behavior regularization, including TD3+BC, ReBRAC, SSAR, and our ATLAS mechanism probe.

TODO: A-line owner: confirm the exact value-conservatism method names and implementation source.

TODO: B-line owner: confirm whether the non-conservative contrast is PPO, SAC, vanilla TD3-style fine-tuning, or a combination.

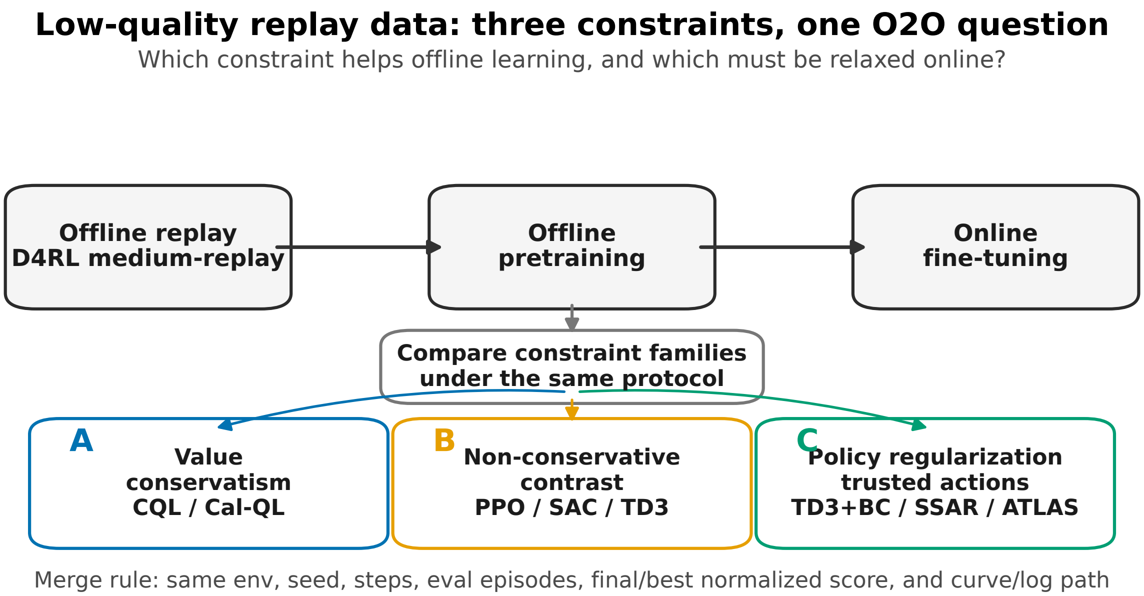

Figure 1: Project framing. The final report compares three constraint families under one offline-to-online question.

The current draft is written before A/B-line results are complete. Therefore, the filled evidence mainly comes from C-line. These results are already informative. On `hopper-medium-replay-v2`, simple TD3+BC is weak, stronger behavior regularization helps, and SSAR is a strong teacher. More importantly, SSAR's advantage appears tied to trusted-action selection rather than generic regularization. This motivates ATLAS, a lightweight selector that distills trusted-action labels from a cached SSAR/IQL-qv teacher into a reusable trust score. The online results then reveal the main tension: a constraint that is useful during offline learning can become harmful if it remains fixed during online fine-tuning.

Our intended contributions are:

1. A small empirical comparison framework for low-quality offline-to-online RL across value conservatism, non-conservative online learning, and policy regularization.

2. Mechanism evidence that trusted-action selection is a high-value component of behavior-regularized offline RL.

3. ATLAS, a C-line mechanism probe showing that aligned teacher trust labels can improve offline learning while shuffled labels fail.

4. A cautionary offline-to-online finding: teacher-label regularization can over-constrain online adaptation if carried forward naively, motivating constraint release as the next decisive test.

TODO: Final editor: after A/B-line results arrive, rewrite contributions so the paper is not C-line only.

2 Background and Setup

2.1 Offline-to-Online RL

In offline-to-online RL, an agent first learns from a fixed dataset $\mathcal{D}$ and then improves through online interaction. We use replay-style D4RL MuJoCo tasks because D4RL was designed to expose offline RL failure modes under mixed-quality logged data and provides a standard normalized score interface [2]. The normalized D4RL score used in this project is:

$$\text{score} = \texttt{env.get\_normalized\_score(raw\_return)} \times 100.$$

2.2 Algorithm Families

Value conservatism. Value-conservative methods penalize or regularize value estimates to avoid overestimating actions outside the data distribution. CQL is the canonical anchor for this family [7]; Cal-QL extends this line toward offline-to-online fine-tuning by calibrating conservative value estimates so they remain useful during online learning [11]. This is the intended A-line.

TODO: A-line owner: fill exact method, source repo or implementation, hyperparameters, and result summary.

Normal or non-conservative contrast. Normal online learners such as PPO, SAC, or vanilla TD3-style fine-tuning provide a contrast against explicitly conservative or behavior-regularized methods [4, 5, 14]. This track is useful even if raw scores are lower, because it answers what happens when explicit conservatism and trusted-action regularization are removed.

TODO: B-line owner: fill exact method, whether it uses offline pretraining, online horizon, and result summary.

Policy and behavior regularization. Behavior-regularized methods constrain policy updates toward dataset actions. TD3+BC is a simple anchor [3], ReBRAC improves the regularization design [15], and SSAR introduces a stronger trusted-action mechanism through selective state-adaptive regularization [10]. PRDC and A2PR are recent policy-regularization neighbors that motivate action-level constraint selection under suboptimal data [9, 13]. ATLAS is our C-line probe for distilling trusted-action information.

2.3 Closest Offline-to-Online Neighbors

The refined claim should not be that online constraint release is completely new. Prior offline-to-online work already studies this stability-efficiency tension. Adaptive BC regularization adjusts the behavior-cloning loss during online fine-tuning [17]; PROTO explicitly relaxes a policy-regularization term during fine-tuning [8]; ENOTO uses Q-ensembles and loosens pessimism for online improvement [16]; and SUF argues for stabilizing unconstrained fine-tuning without retaining offline constraints [1]. Our narrower C-line contribution is a controlled label-quality decomposition: SSAR-style action-level trust labels help offline, shuffled trust weights fail, and fixed teacher-label regularization can obstruct online adaptation.

2.4 Advantage-Weighted Policy Extraction

ATLAS is also adjacent to advantage-weighted supervised policy extraction. AWR and IQL both use value-derived weights to turn policy improvement into a supervised action-fitting problem [6, 12]. The difference is that ATLAS does not estimate its own online advantage during fine-tuning. It distills cached SSAR/IQL-qv trusted-action labels into a selector, so the key ablation is whether the teacher-label alignment itself matters.

2.5 Minimum Comparable Record

Every experiment should report:

- method and method family;
- environment and seed;
- offline steps and online steps;
- evaluation episodes;
- final normalized score and best normalized score;
- curve or log path;
- one-sentence interpretation.

3 C-Line Mechanism: Trusted-Action Selection

The C-line began as a policy-regularization track. Initial smoke experiments showed that plain TD3+BC was a weak anchor on `hopper-medium-replay-v2`, while ReBRAC-lite and SSAR were stronger. This raised a mechanism question: is SSAR stronger because it regularizes more, or because it regularizes toward better-selected actions?

Our evidence points to trusted-action selection. A cheap SSAR variant without IQL-based trusted selection performs much worse than full SSAR. Naive substitutes, including return-ranked trust, online Q-gap trust, and behavior-consistency trust, do not reliably solve the problem. This suggests that the useful signal is not arbitrary weighting; it is the choice of which dataset actions deserve trust.

ATLAS tests whether this trusted-action signal can be distilled. Given a cached SSAR/IQL-qv teacher, ATLAS trains a selector

$$g_\phi(s, a) \in [0, 1],$$

which maps a state-action pair to a trust score. This score is then used as a weight in TD3+BC-style learning.

In the current implementation plan, g_ϕ is a small multilayer perceptron over the concatenated state-action vector, trained with binary cross entropy on teacher-derived trusted-action labels. The selector is not an actor or critic replacement; it only controls how strongly the actor is regularized toward each dataset action.

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_Q + \lambda(t)\, \mathbb{E}_{(s_i,a_i)\sim\mathcal{D}} \left[w_i \|\pi_\theta(s_i) - a_i\|_2^2 \right], \qquad w_i = \text{stopgrad}(\text{clip}(g_\phi(s_i, a_i), 0, 1)).$$

For offline training, $\lambda(t)$ is fixed. For online fine-tuning, the key test is whether this regularization should stay fixed or follow a linear release schedule:

$$\lambda(t) = \lambda_0 \max(0, 1 - t/K).$$

This release schedule is intentionally simple. It tests the observed failure mode directly before adding a learned gate or larger selector.

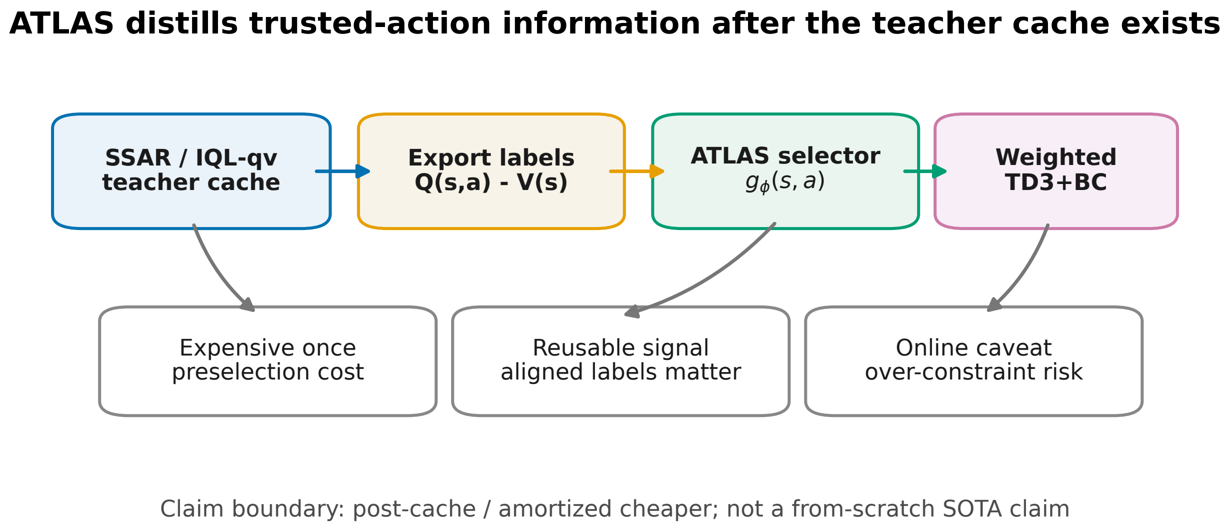

Figure 2: ATLAS mechanism. ATLAS distills trusted-action labels from a cached SSAR/IQL-qv teacher and uses them for weighted TD3+BC-style learning.

TODO: C-line owner: add cost/time numbers: SSAR full IQL-qv preselection vs ATLAS post-cache selector training.

The current claim is intentionally limited. ATLAS is not presented as a new SOTA offline RL algorithm. It is a mechanism probe showing that aligned teacher trust labels matter and can be reused after the expensive teacher cache has been produced.

4 Experiments

4.1 Experimental Questions

- Q1: Under low-quality replay data, do policy/behavior regularization methods outperform a weak TD3+BC anchor?
- Q2: Is SSAR's gain plausibly tied to trusted-action selection rather than generic regularization?
- Q3: Can ATLAS distill the trusted-action signal into a reusable selector?
- Q4: Does the offline advantage persist during online fine-tuning?
- Q5: If fixed teacher regularization hurts online adaptation, does a simple release schedule recover the loss?

TODO: A-line owner: add the value-conservatism experimental question after CQL / Cal-QL results arrive.

TODO: B-line owner: add the non-conservative online contrast question after PPO / SAC / vanilla results arrive.

Method	Seed	Steps	Eval eps	Final	Best	Interpretation
TD3+BC	0	50k	5	22.43	22.43	weak anchor
ReBRAC-lite	0	50k	5	34.48	34.48	stronger regularized baseline
PRDC official	0	50k	5	23.54	23.54	no clear uplift in smoke
A2PR official	0	50k	5	22.31	22.81	no clear uplift in smoke
SSAR full IQL-qv	0	50k	5	38.56	43.97	strongest 50k modern baseline
SSAR cached IQL-qv	0	100k	5	92.44	100.98	high-variance upper anchor
SSAR full IQL-qv	1	100k	20	60.88	99.22	near-100 spike repeats; unstable final
IQL compact	0	100k	5	45.27	81.28	strong value-style baseline
CQL compact	0	50k	5	39.81	39.81	conservative value anchor
ATLAS	0	100k	5	69.97	69.97	positive offline signal
ATLAS	1	100k	20	68.11	68.11	seed stability check
ATLAS shuffled labels	0	50k	5	18.78	19.35	label alignment matters

Table 1: C-line offline and mechanism results on `hopper-medium-replay-v2`.

Method	Seed	Steps	Eval eps	Final	Best	Interpretation
SSAR full IQL-qv	0	100k	10	94.28	94.60	strong teacher anchor
ATLAS	0	100k	10	71.26	77.86	signal survives, but gap remains

Table 2: Second-environment check on `walker2d-medium-replay-v2`.

4.2 Main Offline and Mechanism Results

Environment: `hopper-medium-replay-v2`.

4.3 Second Replay Environment

4.4 Minimal Offline-to-Online Slice

Environment: `hopper-medium-replay-v2`, seed0, 50k offline + 10k online.

TODO: C-line owner: if one more C-line run is launched, prioritize the clean 2x2 release test {TD3+BC, ATLAS} x {fixed, linear-release}, not PRDC/A2PR expansion.

4.5 Cross-Line Fill-In Table

TODO: A-line owner: replace the A row with real method, steps, eval episodes, final/best score, and log path.

TODO: B-line owner: replace the B row with real method, steps, eval episodes, final/best score, and log path.

5 Discussion

The current C-line evidence suggests a distinction between offline stabilization and online adaptability. Trusted-action selection can stabilize learning from low-quality replay data, but the same constraint can become harmful once the agent has access to fresh online data. ATLAS begins from a stronger offline policy than TD3+BC, yet its online fine-tuning score drops below its offline endpoint, while TD3+BC improves during the short online slice.

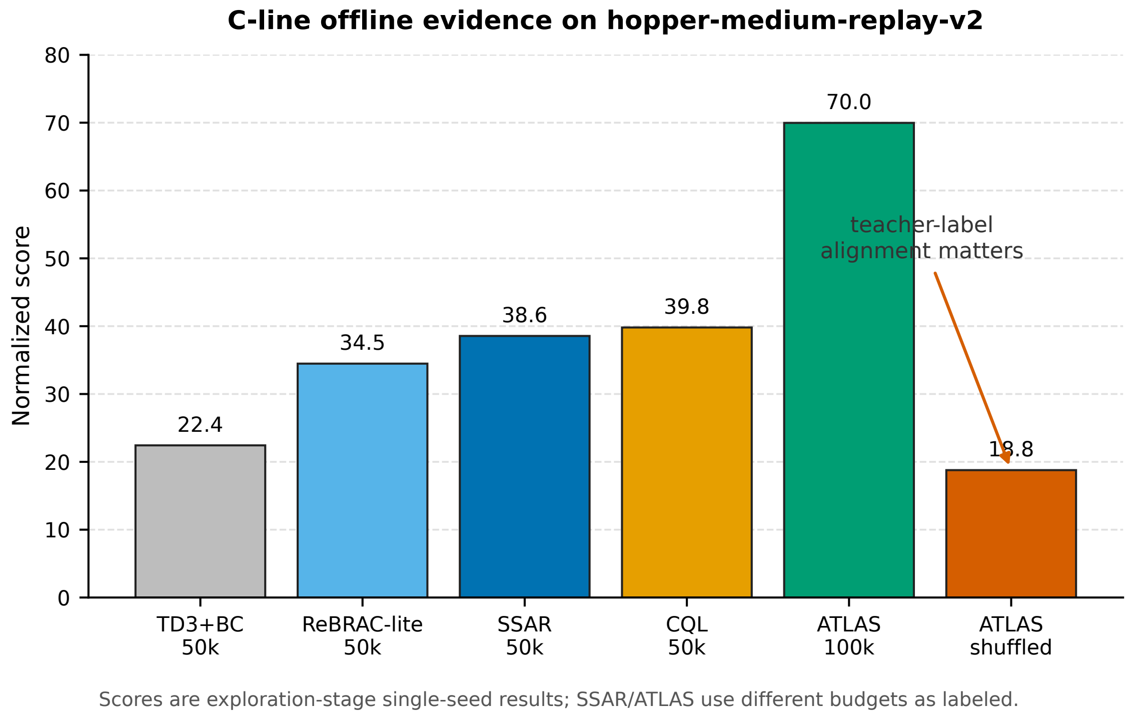

Figure 3: C-line offline evidence on `hopper-medium-replay-v2`. Aligned ATLAS labels improve the offline endpoint, while shuffled labels collapse.

Method	Offline final	Online final	Online best	Interpretation
TD3+BC decay	22.43	39.64	39.64	online fine-tuning works
TD3+BC fixed	22.43	36.78	36.78	online improves but less than decay
ATLAS decay	45.29	37.97	38.26	offline advantage does not persist
ATLAS fixed	45.29	29.22	32.27	fixed teacher regularization over-constrains adaptation
ATLAS decay, online fraction 0.25	45.29	31.46	43.73	transient spike, worse final

Table 3: Minimal offline-to-online smoke.

This argues against a simple "more conservatism is always better" story. A more plausible view is temporal: conservative or trusted-action constraints may help during offline pretraining, but online fine-tuning may require releasing the constraint. The next C-line test should therefore be a minimal release experiment rather than another broad baseline sweep.

TODO: LJY / B-line: if Gate-aware release is kept, position it here as follow-up or optional mechanism, not as a replacement for the main project.

TODO: A-line owner: add whether value conservatism appears stable, over-conservative, or implementation-sensitive.

TODO: B-line owner: add whether non-conservative online learning adapts faster, starts weaker, or remains unstable.

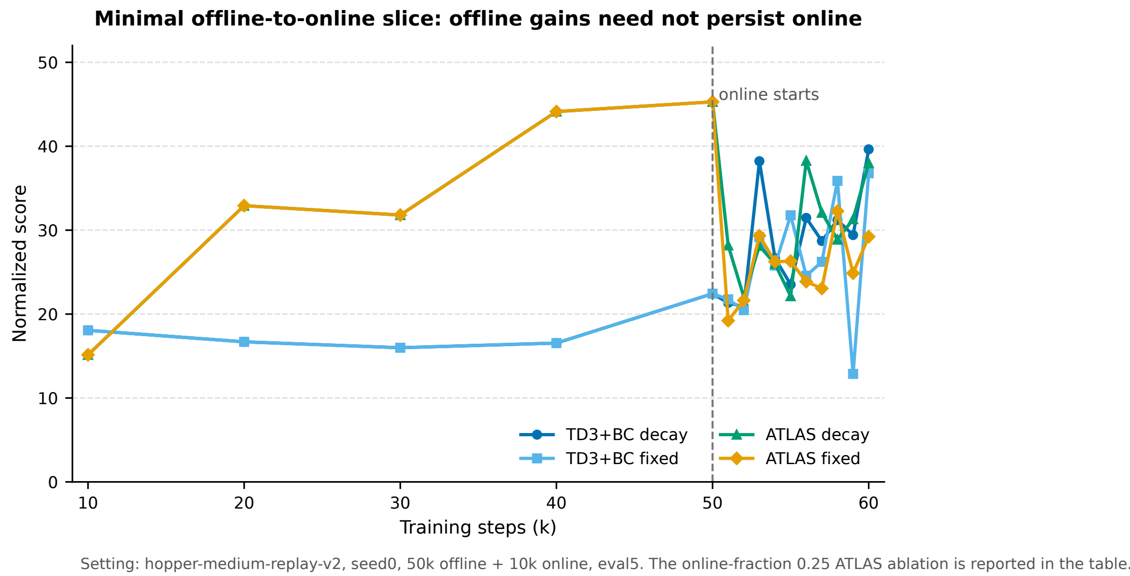

Figure 4: Minimal offline-to-online slice. TD3+BC improves during online fine-tuning, while ATLAS loses its offline advantage under the current online regularization.

Track	Family	Method	Env	Seed	Offline	Online	Eval eps	Final	Best	Status
A	Value conservatism	TODO	hopper-medium-replay-v2	0	TODO	TODO	TODO	TODO	TODO	waiting
B	Non-conservative contrast	TODO	hopper-medium-replay-v2	0	TODO	TODO	TODO	TODO	TODO	waiting
C	Policy regularization	TD3+BC O2O decay	hopper-medium-replay-v2	0	50k	10k	5	39.64	39.64	done
C	Trusted-action selection	ATLAS offline	hopper-medium-replay-v2	0	100k	0	5	69.97	69.97	done
C	Trusted-action selection	ATLAS offline	hopper-medium-replay-v2	1	100k	0	20	68.11	68.11	done

Table 4: Template table for merging A/B/C-line results.

6 Limitations

First, most current results are exploration-stage smoke experiments rather than full multi-seed benchmarks. The goal was to identify high-value mechanisms under budget constraints.

Second, SSAR and ATLAS depend on an expensive IQL-qv teacher cache. ATLAS should be described as post-cache or amortized cheaper, not cheaper from scratch.

Third, the online slice is intentionally short. It demonstrates that online fine-tuning is wired up and exposes a limitation of ATLAS, but it is not a final O2O benchmark.

Fourth, A-line and B-line results are not yet merged. The final cross-line claim should wait until those tracks provide comparable results.

7 Conclusion

This draft studies low-quality offline-to-online RL through value conservatism, non-conservative online learning, and policy/behavior regularization. The filled C-line evidence supports a cautious mechanism finding: trusted-action selection is important for behavior-regularized offline learning, and aligned teacher labels can substantially improve an ATLAS-style selector. At the same time, the first online slice shows that this offline advantage does not automatically transfer to online fine-tuning. The current project direction is therefore not to claim ATLAS as a new SOTA method, but to study when trusted constraints should be used offline and when they should be released online.

TODO: Final editor: rewrite conclusion after A/B-line results arrive. The final conclusion should make a cross-line statement, not only a C-line statement.

References

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A Artifact Index

Primary C-line evidence files:

- `refine-logs/C_LINE_MERGE_PACKAGE_20260513.md`
- `refine-logs/C_LINE_RESULTS_TABLE_20260513.md`
- `refine-logs/remote-results/o2o_minimal_20260513/README.md`
- `refine-logs/NOVELTY_CHECK_20260513.md`
- `refine-logs/EXPERIMENT_TRACKER.md`

Key executable entrypoints:

- `algorithms/td3_bc_o2o.py`
- `scripts/run_o2o_minimal.sh`
- `scripts/export_iql_trust_labels.py`
- `scripts/train_atlas_selector.py`
- `scripts/make_label_control.py`

B Command and Hyperparameter TODOs

B.1 C-Line Reproducibility Commands

The shared local runners are meant to document the protocol rather than hide it behind one-off shell history. The standard C-line smoke uses one diagnostic replay dataset, one seed, 50k offline steps, and five evaluation episodes:

```
ENV=hopper-medium-replay-v2 SEED=0 STEPS=50000 \
RUNS="td3_bc rebrac_lite iql cql" \
bash scripts/run_c_track_smoke.sh
```

ATLAS uses a cached SSAR/IQL-qv teacher. The export step scores every D4RL transition with the teacher:

```
CHECKPOINT=/path/to/ssar_iql_qv_0.7_model.pth
LABELS=/path/to/iql_qv_labels_seed0.npz
python3 scripts/export_iql_trust_labels.py \
  --env hopper-medium-replay-v2 \
  --seed 0 \
  --checkpoint "$CHECKPOINT" \
  --output "$LABELS"
```

The selector is then trained from the exported labels:

```
LABELS=/path/to/iql_qv_labels_seed0.npz
ATLAS=/path/to/atlas_selector_seed0.npz
MODEL=/path/to/atlas_selector_seed0.pt
python3 scripts/train_atlas_selector.py \
  --env hopper-medium-replay-v2 \
  --seed 0 \
  --labels "$LABELS" \
  --target_key hard_trust \
  --output_predictions "$ATLAS" \
  --output_model "$MODEL"
```

Offline ATLAS uses the trained selector as a label file:

```
ATLAS=/path/to/atlas_selector_seed0.npz
ENV=hopper-medium-replay-v2 SEED=0 STEPS=100000 \
RUNS="trusted_td3_bc_label_file" \
LABEL_PATH="$ATLAS" \
LABEL_SCORE_KEY=trust_score \
LABEL_ALGO_NAME=trusted_td3_bc_atlas \
bash scripts/run_c_track_smoke.sh
```

The minimal online slice is run through the O2O runner:

```
ATLAS=/path/to/atlas_selector_seed0.npz
RUN=td3_bc_o2o bash scripts/run_o2o_minimal.sh
RUN=td3_bc_o2o_fixed bash scripts/run_o2o_minimal.sh
RUN=atlas_o2o LABEL_PATH="$ATLAS" bash scripts/run_o2o_minimal.sh
RUN=atlas_o2o_fixed LABEL_PATH="$ATLAS" bash scripts/run_o2o_minimal.sh
```

For the next C-line run, the recommended protocol is not another broad baseline sweep. It is the same runner with a clean fixed-vs-linear-release comparison under one seed and one env.

TODO: A-line owner: add exact command and config for CQL / Cal-QL runs.

TODO: B-line owner: add exact command and config for PPO / SAC / vanilla online runs.